success.

The first part of the Bluetooth standard dictates that all devices connect via a specific set of radio frequencies. The rest of the Bluetooth standard explains the precise protocols used to transmit and receive data over the wireless connection. All Bluetooth devices must use the same protocols, so they all talk the same electronic language.

We have mentioned that Bluetooth transmits data using radio frequency and using a low-frequency portion of the electromagnetic spectrum. TV Stations and FM Radio stations also use that portion of the spectrum.

In the earlier portions of this Fast Track, we've already mentioned how radio signals work and how data can be transmitted wirelessly, so we shall avoid going through it again. Bluetooth uses the 2.40 GHz to 2.48 GHz band. That is, the Bluetooth Radio transmits and receives data at a frequency of 2.40 GHz (gigahertz). Unlike infrared transmissions, which use light waves and require connecting devices to be in sight of each other, radio waves have no line-of-sight requirements and can, in fact, pass through most solid objects. This means that a Bluetooth radio can transmit its RF signals from inside a briefcase, or through office walls. The frequency around the 2.40 GHz band is termed the ISM band (for industrial, scientific and medical purposes). It is free for anyone to use, for any purpose. The benefit of such free licensing is that it allows for experimentation with free communication techniques. The disadvantage is that it leads to crowding of the spectrum since space within the band is finite, and several other devices also use it. Both voice and data communication via Bluetooth utilise this portion of the spectrum.

> Bluetooth doesn't need the devices to be in the line of sight, unlike earlier infrared systems

Currently, the 2.4-GHz band is used by devices such as cordless phones, some urban and suburban wireless communication networks, 802.11 wireless networks and microwave ovens, among others.

So, if you use a Bluetooth device in and around other